

Homework 4 Due Monday March 16, 11:59pm

For each problem, be sure to explain why your controller achieves its objective.

Problem 1 Text problem 10.6.

Problem 2 Text problem 12.1(4). [Hint: Use feedback linearization to construct a state-feedback stabilizer $u = \alpha(x_1, x_2)$. Use the observer structure in (11.17) of Section 11.3 (“Global observers”) to build a state observer $\hat{x}$. Show that the controller $u = \alpha(x_1, \hat{x}_2)$ renders the origin of the overall system (plant + observer) globally asymptotically stable using the third bullet point of Theorem 12.1.]

Problem 3 Text problem 13.11(a).

Problem 4 Text problem 13.12(a).